

Renewable Energy Grid Stability and Battery Storage

Draft 4

1 Introduction

Solar and wind generation are variable, while electricity demand changes throughout the day and across different days and seasons. This variability creates uncertainty about whether the available electricity will be sufficient to meet demand at a given time.

This project develops a probabilistic model of a renewable-energy system with battery storage. The model combines stochastic renewable generation, consumer demand, and battery operation to determine the probability that available power is insufficient to meet demand.

The central random variable of the model is the net energy balance,

$$\boxed{E_t = W_t + S_t + B_t - D_t} \tag{1}$$

where

- W_t = electrical power generated by wind at time t ,
- S_t = electrical power generated by solar at time t ,
- B_t = power supplied by the battery to meet demand at time t ,
- D_t = electricity demand at time t ,
- E_t = net available power at time t .

Renewable generation is therefore

$$\boxed{G_t = W_t + S_t}. \tag{2}$$

A loss-of-load event occurs when

$$E_t < 0. \tag{3}$$

Thus, the main probability of interest is

$$\boxed{P(E_t < 0)}. \tag{4}$$

2 Research Question

Main Research Question

To what extent can battery storage reduce the probability of loss of load in a grid with stochastic renewable generation and consumer demand?

Supporting Questions

1. How can wind generation, solar generation, and electricity demand be represented probabilistically?
2. How does battery capacity and state-of-charge affect the distribution of E_t and therefore $P(E_t < 0)$?
3. How sensitive is the probability of loss of load to the assumptions made about the underlying probability distributions?

3 Overall Model Structure

The model follows the sequence

$$\text{Wind Speed} \rightarrow \text{Wind Power} \quad (5)$$

$$\text{Solar Irradiance} \rightarrow \text{Solar Power} \quad (6)$$

together with consumer demand and battery operation:

$$\boxed{W_t + S_t + B_t - D_t \rightarrow E_t \rightarrow P(E_t < 0)} \quad (7)$$

The underlying environmental variables are wind speed and solar irradiance. These are converted into electrical power before being used in the energy-balance equation.

Let

$$V_t = \text{wind speed at time } t \quad (8)$$

and

$$R_t = \text{solar irradiance at time } t. \quad (9)$$

Then

$$V_t \rightarrow W_t \quad (10)$$

and

$$R_t \rightarrow S_t. \quad (11)$$

4 Wind Generation

Let V_t denote wind speed. As a reference model, wind speed is represented by a Weibull distribution:

$$V_t \sim \text{Weibull}(k, \lambda), \quad (12)$$

with probability density function

$$f_V(v) = \frac{k}{\lambda} \left(\frac{v}{\lambda}\right)^{k-1} \exp \left[- \left(\frac{v}{\lambda}\right)^k \right], \quad v \geq 0. \quad (13)$$

The parameters k and λ will be estimated from the wind-speed data.

Wind speed is not itself electrical power. A turbine power function is therefore required:

$$W_t = f_W(V_t). \quad (14)$$

For the reference model, a simplified turbine power curve is

$$W_t = \begin{cases} 0, & V_t < V_{ci}, \\ P_r \left(\frac{V_t - V_{ci}}{V_r - V_{ci}} \right)^3, & V_{ci} \leq V_t < V_r, \\ P_r, & V_r \leq V_t < V_{co}, \\ 0, & V_t \geq V_{co}, \end{cases} \quad (15)$$

where

- V_{ci} = cut-in wind speed,
- V_r = rated wind speed,
- V_{co} = cut-out wind speed,
- P_r = rated wind power.

This converts the probability distribution of wind speed into a probability distribution of wind power.

5 Solar Generation

Let R_t denote solar irradiance. To model the stochastic component of solar generation, normalized daylight irradiance is represented by a Beta distribution:

$$R_t^* \sim \text{Beta}(\alpha, \beta), \quad 0 \leq R_t^* \leq 1. \quad (16)$$

Its probability density function is

$$f_R(r) = \frac{r^{\alpha-1}(1-r)^{\beta-1}}{B(\alpha, \beta)}, \quad 0 \leq r \leq 1. \quad (17)$$

The normalized irradiance is converted into solar power using

$$S_t = \begin{cases} P_r^{solar} R_t^*, & t \in \text{daylight}, \\ 0, & t \in \text{night}. \end{cases} \quad (18)$$

Thus, solar generation is explicitly constrained to zero during nighttime.

The parameters α and β may depend on the time of day and season so that the model can capture changing solar conditions.

6 Consumer Electricity Demand

Consumer demand is represented using a Normal distribution.

Let

$$D_t = \text{electricity demand at time } t. \quad (19)$$

The reference model assumes

$$\boxed{D_t \sim \mathcal{N}(\mu_t, \sigma_t^2)} \quad (20)$$

where

- μ_t = expected electricity demand at time t ,
- σ_t^2 = variance of electricity demand at time t .

The probability density function is

$$\boxed{f_D(d) = \frac{1}{\sigma_t \sqrt{2\pi}} \exp \left[-\frac{(d - \mu_t)^2}{2\sigma_t^2} \right]}. \quad (21)$$

Therefore, the probability of observing demand near a particular value d is determined by this density.

The parameters μ_t and σ_t are not assumed to be constant throughout the day. They can be estimated separately according to factors such as

$$\text{hour of day,} \quad \text{day type,} \quad \text{season}. \quad (22)$$

Thus,

$$\mu_t = f(\text{hour, day type, season}) \quad (23)$$

and similarly for σ_t .

For example, electricity demand at 3:00 AM and 8:00 PM may have different means and variances.

The Normal model is a reference approximation and will be evaluated against the observed demand data.

7 Battery State-of-Charge Dynamics

The battery must be distinguished from the power it supplies.

Let

$$C_t = \text{energy stored in the battery at time } t \quad (24)$$

and

$$C_{\max} = \text{maximum battery energy capacity.} \quad (25)$$

The stored energy is measured in kWh, whereas battery power is measured in kW.

We use a signed battery power variable $P_{BESS}(t)$:

$$P_{BESS}(t) \begin{cases} < 0, & \text{battery is charging,} \\ > 0, & \text{battery is discharging.} \end{cases} \quad (26)$$

The state of charge can then be represented continuously as

$$C(t) = C(t_0) - \int_{t_0}^t \left[\eta_{ch} P_{BESS}^-(\tau) + \frac{1}{\eta_{dis}} P_{BESS}^+(\tau) \right] d\tau \quad (27)$$

where

$$P_{BESS}^-(t) = \min(P_{BESS}(t), 0) \quad (28)$$

and

$$P_{BESS}^+(t) = \max(P_{BESS}(t), 0). \quad (29)$$

Because $P_{BESS}^-(t)$ is negative during charging, the first term increases the stored energy. During discharging, the second term reduces the stored energy.

Here,

- $C_{\max}$ = maximum battery energy capacity,
- η_{ch} = charging efficiency,
- η_{dis} = discharging efficiency,
- P_{BESS}^- = charging power,
- P_{BESS}^+ = discharging power.

7.1 Discrete-Time Battery Model

Since the renewable generation and demand are modelled at discrete time intervals, the battery equation can be written more conveniently using Δt .

Let

$$C_t = \text{stored battery energy at the beginning of interval } t. \quad (30)$$

If the battery is charging,

$$C_{t+1} = C_t + \eta_{ch} P_t^{ch} \Delta t. \quad (31)$$

If the battery is discharging,

$$C_{t+1} = C_t - \frac{P_t^{dis} \Delta t}{\eta_{dis}}. \quad (32)$$

The battery must satisfy

$$\boxed{C_{\min} \leq C_t \leq C_{\max}}. \quad (33)$$

The battery power must also satisfy

$$\boxed{-P_{ch,\max} \leq P_{BESS}(t) \leq P_{dis,\max}}. \quad (34)$$

7.2 Charging

The renewable generation is

$$G_t = W_t + S_t. \quad (35)$$

When

$$G_t > D_t, \quad (36)$$

there is surplus renewable power

$$G_t - D_t. \quad (37)$$

The charging power is therefore bounded by the available surplus, the maximum charging power, and the remaining battery capacity:

$$P_t^{ch} = \min \left(G_t - D_t, P_{ch,\max}, \frac{C_{\max} - C_t}{\eta_{ch} \Delta t} \right). \quad (38)$$

The battery state then becomes

$$\boxed{C_{t+1} = C_t + \eta_{ch} P_t^{ch} \Delta t}. \quad (39)$$

7.3 Discharging

When

$$G_t < D_t, \quad (40)$$

the renewable-generation deficit is

$$D_t - G_t. \quad (41)$$

The battery can supply part of this deficit.

Let

$$B_t = P_t^{dis} \quad (42)$$

be the power supplied by the battery to meet demand.

Then

$$P_t^{dis} = \min \left(D_t - G_t, P_{dis,max}, \frac{\eta_{dis} C_t}{\Delta t} \right). \quad (43)$$

The battery state becomes

$$\boxed{C_{t+1} = C_t - \frac{P_t^{dis} \Delta t}{\eta_{dis}}}. \quad (44)$$

Therefore,

$$\boxed{B_t = P_t^{dis} \geq 0} \quad (45)$$

is the battery power that actually contributes to meeting demand.

This distinction is important:

$$\boxed{C_t \neq B_t} \quad (46)$$

because C_t represents stored energy, while B_t represents the power supplied during the current time interval.

8 Net Energy Balance

After combining renewable generation, battery support, and demand, the net energy balance is

$$\boxed{E_t = W_t + S_t + B_t - D_t}. \quad (47)$$

Since

$$G_t = W_t + S_t, \quad (48)$$

this can also be written as

$$E_t = G_t + B_t - D_t. \quad (49)$$

The three possible cases are

$$\begin{cases} E_t > 0 & \text{surplus available power,} \\ E_t = 0 & \text{demand is exactly met,} \\ E_t < 0 & \text{insufficient available power.} \end{cases} \quad (50)$$

Thus, a loss-of-load event occurs when

$$\boxed{E_t < 0}. \quad (51)$$

Equivalently,

$$D_t > W_t + S_t + B_t. \quad (52)$$

9 Probability Distribution of Net Energy

The main random variable of interest is

$$E_t = W_t + S_t + B_t - D_t. \quad (53)$$

For the initial analytical reference model, we assume that the component random variables are independent at a given time t :

$$\boxed{W_t \perp S_t \perp B_t \perp D_t}. \quad (54)$$

This is a simplifying assumption used to derive the reference distribution. The battery's actual state evolves through time and can therefore introduce temporal dependence, which can be investigated separately.

Let f_W , f_S , f_B , and f_D denote the probability density functions of wind power, solar power, battery output, and demand.

For a particular value e of the net energy balance,

$$E_t = e \quad (55)$$

implies

$$d = w + s + b - e. \quad (56)$$

Therefore, under the independence assumption, the probability density function of E_t is

$$\boxed{f_E(e) = \iiint f_W(w) f_S(s) f_B(b) f_D(w + s + b - e) dw ds db}. \quad (57)$$

Since the demand distribution is Normal,

$$f_D(d) = \frac{1}{\sigma_t \sqrt{2\pi}} \exp \left[-\frac{(d - \mu_t)^2}{2\sigma_t^2} \right]. \quad (58)$$

Therefore,

$$f_E(e) = \iiint f_W(w) f_S(s) f_B(b) \frac{1}{\sigma_t \sqrt{2\pi}} \exp \left[-\frac{(w + s + b - e - \mu_t)^2}{2\sigma_t^2} \right] dw ds db. \quad (59)$$

This shows explicitly how the selected probability distributions determine the distribution of the final random variable E_t .

10 Probability of Loss of Load

The probability of a loss-of-load event is

$$P(E_t < 0). \quad (60)$$

Once $f_E(e)$ is known,

$$P(E_t < 0) = \int_{-\infty}^0 f_E(e) de. \quad (61)$$

This integral represents all probability associated with the negative values of the net energy distribution.

The same event can be expressed as

$$E_t < 0 \iff D_t > W_t + S_t + B_t. \quad (62)$$

Therefore,

$$P(E_t < 0) = P(D_t > W_t + S_t + B_t). \quad (63)$$

For the Normal demand model,

$$P(D_t > x) = 1 - \Phi \left(\frac{x - \mu_t}{\sigma_t} \right), \quad (64)$$

where Φ is the standard Normal cumulative distribution function.

Hence, under the independence assumption,

$$P(E_t < 0) = \iiint \left[1 - \Phi \left(\frac{w + s + b - \mu_t}{\sigma_t} \right) \right] f_W(w) f_S(s) f_B(b) dw ds db. \quad (65)$$

This provides a direct mathematical expression for the probability that demand exceeds available renewable generation and battery support.

11 Deficit and Indicator Variable

The magnitude of an energy deficit is

$$\boxed{\text{Deficit}_t = \max(0, -E_t)}. \quad (66)$$

An indicator variable can be used to identify loss-of-load events:

$$I_t = \begin{cases} 1, & E_t < 0, \\ 0, & E_t \geq 0. \end{cases} \quad (67)$$

Therefore,

$$P(I_t = 1) = P(E_t < 0). \quad (68)$$

For T observed time intervals, the empirical Loss-of-Load Probability is

$$\boxed{\widehat{LOLP} = \frac{1}{T} \sum_{t=1}^T I_t}. \quad (69)$$

12 Data and Model Fitting

The model will be applied to real data.

The required data are

- wind speed with timestamps,
- solar irradiance with timestamps,
- electricity demand with timestamps,
- information about date, time, and day type where available.

The analysis will proceed as follows:

1. Obtain and clean wind, solar, and demand data.
2. Fit a Weibull distribution to wind-speed data.
3. Fit a Beta distribution to normalized daylight solar irradiance.
4. Estimate μ_t and σ_t for the Normal demand model from the electricity-demand data.
5. Convert wind speed and solar irradiance into electrical generation.
6. Simulate battery charging and discharging using the SoC equations.
7. Construct

$$E_t = W_t + S_t + B_t - D_t.$$

8. Estimate the distribution of E_t .

9. Calculate

$$P(E_t < 0).$$

10. Repeat the analysis for different battery capacities.

13 Battery Capacity and Reliability

The main prediction is that increasing battery capacity should generally reduce the probability of loss of load:

$$\boxed{C_{\max} \uparrow \Rightarrow P(E_t < 0) \downarrow}. \quad (70)$$

However, increasing capacity is expected to have diminishing returns once the battery is sufficiently large to cover most renewable generation deficits.

The relationship will therefore be investigated using

$$P(E_t < 0 \mid C_{\max} = C) \quad (71)$$

for several values of battery capacity C .

14 Dependence and Markov Extension

The independence assumption is useful for deriving the reference distribution of E_t , but the physical battery system has temporal dependence.

In particular,

$$C_{t+1} = f(C_t, G_t, D_t). \quad (72)$$

Therefore, the amount of battery power available at time $t + 1$ depends partly on what happened during previous time intervals.

If the battery state is discretized into states $i, j, \dots$, this can be represented using transition probabilities:

$$P(C_{t+1} = j \mid C_t = i). \quad (73)$$

This provides a possible Markov-chain extension of the reference model.

Similarly, if electricity demand shows significant dependence on its recent values, a temporal demand model could be introduced through

$$P(D_{t+1} \mid D_t). \quad (74)$$

These extensions will only be introduced if they are supported by the data.

15 Assumptions

The reference model makes the following assumptions:

1. Wind speed follows a Weibull distribution.
2. Wind speed is converted to electrical power using a specified turbine power curve.
3. Normalized daylight solar irradiance follows a Beta distribution.
4. Solar generation is zero during nighttime.
5. Electricity demand follows a Normal distribution

$$D_t \sim \mathcal{N}(\mu_t, \sigma_t^2).$$

6. The mean and variance of demand may vary with time of day, day type, and season.
7. Wind power, solar power, battery output, and demand are initially assumed to be independent for the analytical derivation of $f_E(e)$.
8. Battery state-of-charge evolves according to charging, discharging, and efficiency constraints.
9. Battery energy and battery power are treated as separate quantities.
10. The reference time interval is $\Delta t = 1$ hour, unless the selected dataset supports a different resolution.

16 Expected Outcome

The model aims to obtain the probability distribution

$$f_E(e) \tag{75}$$

of the net energy balance and, most importantly, the probability

$$\boxed{P(E_t < 0)}. \tag{76}$$

The effect of battery storage can then be quantified by comparing

$$P(E_t < 0 \mid C_{\max} = C_1), \quad P(E_t < 0 \mid C_{\max} = C_2), \quad \dots \tag{77}$$

for different battery capacities.

The overall model can therefore be summarized as

$$\boxed{\text{Wind} + \text{Solar} + \text{Battery} - \text{Demand} \rightarrow E_t \rightarrow P(E_t < 0)} \tag{78}$$

The final objective is to determine how battery capacity changes the probability and magnitude of insufficient electricity supply.

17 Scope

This project focuses on probabilistic supply-demand reliability. It does not attempt to model complete physical power-grid stability, including voltage stability, frequency dynamics, transmission constraints, or generator faults.

The reference model will first be fitted to real data. More complex dependence structures, such as Markov models for battery state and temporal dependence in demand, can then be introduced as extensions if they improve the model.